

```
/*obstacle avoiding, Bluetooth control, voice control robot car.
```

```
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```

```
*/
```

```
#include <Servo.h>
```

```
#include <AFMotor.h>
```

```
#define Echo A0
```

```
#define Trig A1
```

```
#define motor 10
```

```
#define Speed 100
```

```
#define spoint 103
```

```
char value;
```

```
int distance;
```

```
int Left;
```

```
int Right;
```

```
int L = 0;
```

```
int R = 0;
```

```
int L1 = 0;
```

```
int R1 = 0;
```

```
Servo servo;
```

```
AF_DCMotor M1(1);
```

```
AF_DCMotor M2(2);
```

```
AF_DCMotor M3(3);
```

```
AF_DCMotor M4(4);
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    pinMode(Trig, OUTPUT);
```

```
    pinMode(Echo, INPUT);
```

```
    servo.attach(motor);
```

```
    M1.setSpeed(Speed);
```

```
    M2.setSpeed(Speed);
```

```
}
```

```
void loop() {
```

```
    //Obstacle();
```

```
    Bluetoothcontrol();
```

```
    voicecontrol();
```

```
}
```

```
void Bluetoothcontrol() {
```

```
    if (Serial.available() > 0) {
```

```
        value = Serial.read();
```

```
        Serial.println(value);
```

```
    }
```

```
    if (value == 'F') {
```

```
        forward();
```

```
    } else if (value == 'B') {
```

```
        backward();
```

```
    } else if (value == 'L') {
```

```
        left();
```

```
    } else if (value == 'R') {
```

```
        right();
```

```
    } else if (value == 'S') {
```

```
        Stop();
```

```
    }
```

```
}
```

```
void Obstacle() {
```

```
    distance = ultrasonic();
```

```
    if (distance <= 12) {
```

```
        Stop();
```

```
        backward();
```

```
        delay(100);
```

```
        Stop();
```

```
        L = leftsee();
```

```
        servo.write(spoint);
```

```
        delay(800);
```

```

R = rightsee();
servo.write(spoint);
if (L < R) {
    right();
    delay(500);
    Stop();
    delay(200);
} else if (L > R) {
    left();
    delay(500);
    Stop();
    delay(200);
}
} else {
    forward();
}
}

void voicecontrol() {
    distance = ultrasonic();
    if (distance <= 12) {
        Stop();

    } else {
        if (Serial.available() > 0) {
            value = Serial.read();
            Serial.println(value);
            if (value == '^') {
                forward();
            }
            else if (value == '-') {
                backward();
            }
            else if (value == '<') {
                L = leftsee();
                servo.write(spoint);
                if (L >= 10 ) {
                    left();
                    delay(500);
                    Stop();
                } else if (L < 10) {
                    Stop();
                }
            }
            else if (value == '>') {
                R = rightsee();
                servo.write(spoint);
                if (R >= 10 ) {
                    right();
                    delay(500);
                    Stop();
                } else if (R < 10) {
                    Stop();
                }
            }
            else if (value == '*') {
                Stop();
            }
        }
    }
}

}

// Ultrasonic sensor distance reading function
int ultrasonic() {
    digitalWrite(Trig, LOW);

```

```

    delayMicroseconds(4);
    digitalWrite(Trig, HIGH);
    delayMicroseconds(10);
    digitalWrite(Trig, LOW);
    long t = pulseIn(Echo, HIGH);
    long cm = 0.017 * t; //time convert distance
    return cm;
}

void forward() {
    M1.run(FORWARD);
    M2.run(FORWARD);
}

void backward() {
    M1.run(BACKWARD);
    M2.run(BACKWARD);
}

void right() {
    M1.run(FORWARD);
    M2.run(BACKWARD);
}

void left() {
    M1.run(BACKWARD);
    M2.run(FORWARD);
}

void Stop() {
    M1.run(RELEASE);
    M2.run(RELEASE);
}

int rightsee() {
    servo.write(20);
    delay(800);
    Left = ultrasonic();
    return Left;
}

int leftsee() {
    servo.write(180);
    delay(800);
    Right = ultrasonic();
    return Right;
}

```